Electrical storm — Q1

Define electrical storm

- ≥ 3 separate episodes of sustained VT/VF (or appropriate ICD therapies) within 24 hours
- Episodes separated by ≥ 5 minutes of sinus rhythm; incessant VT is a variant
- Associated with high short-term mortality; a medical emergency

Outline your immediate management in the first hour

- Defibrillate/cardiovert unstable VT; check ICD interrogation (appropriate vs inappropriate shocks)
- Treat triggers: replace K+ to 4.5–5 and Mg2+ (2 g IV), stop proarrhythmic drugs, look for ischaemia (ECG, troponin, urgent angiography if ACS)
- Sedation and analgesia (e.g., midazolam/propofol, fentanyl) to blunt sympathetic drive; intubate if needed
- Non-selective β-blocker (propranolol 40 mg 6-hourly orally or esmolol infusion) – the single most effective drug to blunt sympathetic drive
- Amiodarone 150–300 mg IV load then infusion; lidocaine as add-on, especially if ischaemic VT
- Reprogram ICD to avoid repeated painful shocks (ATP first); avoid unnecessary shocks

Electrical storm — Q1 (cont.)

Explain why sympathetic blockade is central to its treatment and how propranolol differs from metoprolol in this setting

- Each shock causes pain/catecholamine surge → more VT (vicious circle)
- Catecholamines shorten refractoriness, increase automaticity and triggered activity
- Propranolol is non-selective (β1 + β2) and lipophilic → central sympatholytic effect
- Propranolol + amiodarone superior to metoprolol + amiodarone in storm (SHIELD/Chatzidou data)

List the options if the storm persists despite drugs

- Deep sedation with intubation and ventilation
- Left stellate ganglion block (LSGB) or thoracic epidural anaesthesia for cardiac sympathetic denervation
- Urgent catheter ablation of the VT substrate (reduces recurrence)
- Mechanical circulatory support (IABP, Impella, VA-ECMO) as bridge if haemodynamically unstable
- Overdrive pacing; surgical cardiac sympathetic denervation; transplant/LVAD evaluation

Electrical storm — Q2

Describe the role of the autonomic nervous system in ventricular arrhythmia and the rationale for neuromodulation

- Cardiac sympathetic fibres pass through the stellate (cervicothoracic) ganglia, mainly left-sided to the ventricles
- Sympathetic activation increases dispersion of repolarisation, triggered activity and re-entry
- Blocking the stellate ganglion removes this input without negative inotropy of large β-blocker doses
- Useful as a rescue or bridge to ablation in refractory storm

Describe how a stellate ganglion block is done and its complications

- Ultrasound-guided injection of local anaesthetic (e.g., bupivacaine/lidocaine) near the stellate ganglion at C6–C7, left side first
- Can be a single shot or catheter infusion; bilateral if left alone fails
- Expected Horner syndrome confirms block
- Complications: vascular injection (vertebral artery → seizures), recurrent laryngeal nerve palsy, pneumothorax, haematoma, phrenic palsy

Electrical storm — Q2 (cont.)

What further therapies would you consider, and what are your goals for this patient over the next 24 hours

- Thoracic epidural (T1–T4) as alternative neuromodulation
- Urgent electrophysiology for mapping and ablation (with MCS support if needed)
- VA-ECMO or Impella if haemodynamic collapse with recurrent VT
- Correct electrolytes; exclude myocarditis (MRI/biopsy) and drug toxicity
- Polymorphic VT with normal QT → look for ischaemia; consider quinidine in selected cases
- Early discussion of advanced heart failure options (LVAD/transplant)

Pulmonary embolism echo (McConnell sign) — Q3

Name the echo sign described and list four other echo findings of acute RV pressure overload

- McConnell sign: RV free-wall akinesia/hypokinesia with apical sparing (normal or hyperkinetic apex)
- RV dilatation: RV/LV basal diameter ratio > 1.0
- Septal flattening / D-shaped LV (paradoxical septal motion)
- Reduced TAPSE (< 16 mm)
- 60/60 sign: RV outflow acceleration time < 60 ms with TR gradient < 60 mmHg
- Dilated non-collapsing IVC; mobile right heart thrombus

Pulmonary embolism echo (McConnell sign) — Q3 (cont.)

Classify the severity of her PE according to the ESC 2019 risk stratification

- High risk = haemodynamic instability (cardiac arrest, SBP < 90 mmHg or vasopressor need, persistent hypotension)
- Intermediate-high = no shock + PESI III–V/sPESI ≥ 1 + RV dysfunction on imaging AND raised troponin
- Intermediate-low = one or none of RV dysfunction/troponin
- She is normotensive with RV dysfunction and raised troponin → intermediate-high risk

Outline her management

- Anticoagulation with LMWH or UFH (UFH if reperfusion may be needed)
- Do not give routine full-dose systemic thrombolysis (PEITHO: less decompensation but more major/intracranial bleeding)
- Monitor closely; rescue thrombolysis (alteplase 100 mg over 2 h) if she deteriorates
- Catheter-directed therapy or surgical embolectomy if decompensating with high bleeding risk
- Careful oxygen; avoid large fluid boluses and intubation if possible (RV collapse)
- PE response team; consider IVC filter only if anticoagulation contraindicated

Pulmonary embolism echo (McConnell sign) — Q4

Explain the pathophysiology of RV failure in massive PE

- Sudden rise in pulmonary vascular resistance (clot + vasoconstriction from serotonin/thromboxane)
- Thin-walled RV dilates, wall tension rises, RV O2 demand increases
- Septum shifts left (ventricular interdependence) → LV under-filling, low cardiac output
- Hypotension lowers RV coronary perfusion → RV ischaemia → spiral of failure

Describe the management of this high-risk PE, including thrombolysis dosing and contraindications

- UFH bolus 80 U/kg then infusion
- Systemic thrombolysis: alteplase 100 mg over 2 h (or 0.6 mg/kg over 15 min, max 50 mg, in peri-arrest)
- Absolute contraindications: prior haemorrhagic stroke, ischaemic stroke < 6 months, CNS tumour, major trauma/surgery/head injury < 3 weeks, active bleeding
- Noradrenaline for hypotension; avoid fluid overload (≤ 500 mL); dobutamine if low output
- Surgical embolectomy or catheter therapy if thrombolysis contraindicated or fails
- VA-ECMO for refractory shock or cardiac arrest

What are the risks of intubating this patient and how would you reduce them

- Induction agents cause vasodilatation and myocardial depression → arrest
- Positive pressure raises RV afterload and reduces venous return
- Prefer awake/HFNC strategy if possible; if needed, ketamine/etomidate, vasopressor running
- Low tidal volume, low PEEP, avoid hypercapnia and hypoxaemia (increase PVR)

Cardiac tamponade and pulsus paradoxus — Q5

Define pulsus paradoxus and explain its mechanism

- Fall in systolic BP > 10 mmHg during inspiration
- Inspiration increases venous return to the RV; in tamponade the fixed pericardial volume makes the septum bulge into the LV
- LV filling and stroke volume fall (exaggerated ventricular interdependence)
- Pulmonary venous pooling on inspiration further reduces LV preload
- Other causes: severe asthma/COPD, massive PE, hypovolaemia, RV infarction

Describe the echocardiographic signs of tamponade

- Pericardial effusion (often circumferential)
- RA systolic collapse (early, sensitive) and RV diastolic collapse (more specific)
- Dilated plethoric IVC (> 2.1 cm, < 50% collapse)
- Respiratory variation of mitral inflow E velocity > 25% (tricuspid > 40%)
- Swinging heart (corresponds to electrical alternans)

Cardiac tamponade and pulsus paradoxus — Q5 (cont.)

Outline her management

- Urgent echo-guided pericardiocentesis (subxiphoid or apical), leave drain
- Supportive: fluid bolus to keep preload, avoid positive-pressure ventilation and sedation if possible
- Vasopressor/inotrope as a temporising bridge only
- Send fluid for cytology, culture, protein, LDH
- Malignant effusion: consider pericardial window or intrapericardial therapy to prevent reaccumulation
- Avoid diuretics and vasodilators

Cardiac tamponade and pulsus paradoxus — Q6

What is the best management of the tamponade in this patient, and why

- Immediate transfer to theatre for emergency surgical repair
- Avoid routine full pericardiocentesis: restoring BP can re-open the tear, extend the dissection and cause fatal rebleeding
- Controlled drainage of small volumes (just enough to keep MAP ~60 mmHg) only if surgery cannot happen at once and patient is peri-arrest

List the classification systems for aortic dissection and the complications of type A

- Stanford A = ascending aorta involved (surgical); B = not involving ascending aorta
- DeBakey I (ascending + beyond), II (ascending only), III (descending only)
- Type A complications: tamponade, acute aortic regurgitation, coronary occlusion (MI, usually RCA), stroke
- Malperfusion of gut, kidneys or limbs; rupture

Describe the medical management of a haemodynamically stable acute dissection

- Analgesia (opioids) to reduce sympathetic drive
- IV β-blocker first (esmolol or labetalol) to HR < 60/min – reduces dP/dt (shear stress)
- Then vasodilator if needed (nitroprusside, nicardipine, GTN) to SBP 100–120 mmHg
- Do not give vasodilator before β-blocker (reflex tachycardia)
- Arterial line (right radial usually), urine output, serial lactate for malperfusion
- Type B: TEVAR if complicated (malperfusion, rupture, refractory pain/hypertension)

VV ECMO recirculation — Q7

Define recirculation and explain why it occurs here

- Oxygenated return blood is drawn back into the drainage cannula without passing through the systemic circulation
- Effective ECMO support falls even though circuit flow is high
- Risk factors: drainage and return cannula tips too close, high pump flow, low cardiac output, raised intrathoracic/abdominal pressure
- High pre-oxygenator saturation (86%) with falling SaO2 strongly suggests recirculation

How would you confirm and quantify it

- Recirculation fraction R = (SpreO2 − SvO2) / (SpostO2 − SvO2) × 100
- SvO2 measured from a true mixed/central venous sample (e.g., PA catheter) – often estimated
- Echo (or X-ray) to check cannula tip positions and distance (ideally > 10 cm apart)
- Ultrasound dilution (Transonic) gives direct measurement
- Clue: rising SpreO2 when flow is increased, with no SaO2 improvement

VV ECMO recirculation — Q7 (cont.)

How would you manage it, and what are the other causes of hypoxaemia on VV ECMO

- Reposition cannulas to increase tip separation (under echo/fluoro)
- Reduce pump flow to find the optimal flow (less recirculation)
- Change configuration (e.g., bicaval dual-lumen cannula with return jet aimed at the tricuspid valve)
- Optimise cardiac output and reduce intrathoracic pressure
- Other causes: ECMO flow too low relative to cardiac output (high native CO), oxygenator failure (check post-oxygenator PO2), low Hb, high VO2 (fever, shivering)
- Treat high native output: cool, sedate, β-blocker in selected cases; add a second drainage cannula

VV ECMO recirculation — Q8

State the EOLIA entry criteria for VV ECMO and whether she meets them

- PaO2/FiO2 < 50 mmHg for > 3 h
- PaO2/FiO2 < 80 mmHg for > 6 h
- pH < 7.25 with PaCO2 ≥ 60 mmHg for > 6 h (RR ↑ to 35, Pplat ≤ 32)
- She meets criteria (P/F 52, pH 7.18/PaCO2 82) after optimal conventional therapy

Describe the ventilator strategy once ECMO is running

- Ultra-protective ventilation ('lung rest'): Pplat ≤ 24–25 cmH2O, driving pressure ≤ 14–15
- Low RR (~10/min), PEEP ≥ 10 cmH2O to prevent atelectasis
- FiO2 reduced to lowest tolerated (e.g., 0.3–0.5)
- Control PaCO2 with sweep gas; avoid rapid PaCO2 fall (> 50% in 24 h, risk of brain injury)

List the main complications of VV ECMO and how anticoagulation is managed

- Bleeding (ICH, cannula site, GI) – most common
- Thrombosis: oxygenator, circuit, cannula-associated DVT
- Haemolysis (plasma free Hb), HIT, acquired von Willebrand syndrome
- Infection, limb ischaemia (femoral), air embolism, cannula dislodgement
- UFH titrated to anti-Xa 0.2–0.3 IU/mL or aPTT 1.5×; lower or stop if bleeding (circuits tolerate brief heparin-free periods)
- Bivalirudin if HIT

North–south syndrome on VA ECMO — Q9

Explain what is happening

- North–south (Harlequin) syndrome = differential hypoxaemia
- Recovering LV ejects deoxygenated blood (poorly oxygenated by oedematous lungs) into the aortic root
- Retrograde ECMO flow from the femoral artery meets native flow at a 'mixing point' in the aorta
- Coronary and cerebral (upper body) circulation get hypoxic blood while lower body gets ECMO blood

How would you monitor for it

- ABG from the right radial artery (closest to coronary/cerebral flow); right-hand SpO2
- Bilateral cerebral NIRS
- Echo to assess aortic valve opening and LV ejection
- Compare right radial PaO2 with post-oxygenator PO2

Outline the management options

- Optimise lung function: increase PEEP and FiO2, treat pulmonary oedema (diuresis/ultrafiltration), bronchoscopy
- Increase ECMO flow to push the mixing point proximally (may increase LV afterload)
- Convert to V-AV ECMO (add return cannula into the internal jugular) or V-V if heart recovered
- Move arterial return to axillary/subclavian or central aorta
- Consider weaning trial if LV has recovered sufficiently

Board 2026 recall · North–south syndrome on VA ECMO · ELSO Interim Guidelines for VA ECMO (Lorusso R, ASAIO J 2021)

North–south syndrome on VA ECMO — Q10

What is the problem, and why does it occur

- LV distension (LV stasis) on VA-ECMO
- Retrograde arterial flow increases LV afterload
- Poorly contracting LV cannot open the aortic valve; bronchial and Thebesian venous return keep filling it
- Results: raised LVEDP → pulmonary oedema, LV/aortic root thrombus, impaired recovery

Describe the strategies for LV unloading (venting)

- Reduce ECMO flow to the minimum that keeps perfusion; add inotrope (dobutamine/milrinone) to help ejection
- IABP (reduces afterload and promotes aortic valve opening)
- Impella ('ECPELLA') – active LV unloading
- Atrial septostomy (balloon) or surgical LV/LA vent
- Monitor with echo, pulse pressure and pulmonary artery pressure

How would you assess readiness to wean from VA-ECMO

- Underlying cause recovering; MAP > 65 with low-dose inotrope; pulsatile arterial trace
- Flow reduction trial to 1–1.5 L/min with echo: LVEF > 20–25%, LVOT VTI ≥ 10 cm, TDSa ≥ 6 cm/s
- Stable lactate and oxygenation during trial; adequate lung function
- If no recovery: bridge to durable LVAD/transplant or palliation (bridge-to-decision)

IABP timing errors — Q11

Describe the normal timing of inflation and deflation and their haemodynamic effects

- Inflation at the dicrotic notch (aortic valve closure, start of diastole)
- Inflation raises diastolic pressure → increases coronary perfusion (diastolic augmentation)
- Deflation just before systole (before the upstroke)
- Deflation lowers end-diastolic aortic pressure → reduced afterload and myocardial O2 demand

Describe the four timing errors, how each appears, and its consequence

- Early inflation: before the dicrotic notch → premature aortic valve closure, raised LVEDP, reduced stroke volume, increased O2 demand
- Late inflation: after the dicrotic notch → sub-optimal diastolic augmentation and coronary perfusion
- Early deflation: augmented pressure falls sharply then a U-shape before systole → poor afterload reduction, possible retrograde coronary/carotid flow
- Late deflation: assisted end-diastolic pressure ≥ unassisted; balloon still inflated at systole → increased afterload and O2 demand (most dangerous)

What are the triggering options

- ECG (R wave) trigger – most common
- Arterial pressure trigger (if ECG poor)
- Pacing spike trigger
- Internal/asynchronous mode only during cardiac arrest/no output

Board 2026 recall · IABP timing errors · Krishna M, Zacharowski K. Principles of intra-aortic balloon pump counterpulsation. Contin Educ Anaesth Crit Care Pain 2009

IABP timing errors — Q12

List the indications and contraindications for IABP

- Indications: mechanical MI complications (VSD, acute MR) as bridge, refractory ischaemia, high-risk PCI/CABG support
- Cardiogenic shock is no longer a routine indication (IABP-SHOCK II: no mortality benefit)
- Absolute contraindications: moderate–severe aortic regurgitation, aortic dissection
- Relative: aortic aneurysm, severe peripheral vascular disease, uncontrolled sepsis/bleeding

Describe the complications of IABP

- Limb ischaemia (check distal pulses); vascular injury, dissection
- Bleeding, thrombocytopenia, haemolysis
- Balloon rupture (blood in the helium line) → gas embolism
- Malposition: renal artery occlusion (too low) or left subclavian/cerebral (too high); tip 2–3 cm below the left subclavian (2nd–3rd rib space on CXR)

Explain why the IABP helps a patient with VSD or acute MR, and what is its effect on cardiac output

- Afterload reduction lowers systemic resistance → more forward flow and less left-to-right shunt/regurgitant volume
- Diastolic augmentation improves coronary perfusion of ischaemic myocardium
- Cardiac output increases modestly (~0.5–1 L/min)
- Needs some native cardiac function and a stable rhythm
- Impella or VA-ECMO if more support needed

Anaphylaxis on beta-blockers — Q13

Why is his anaphylaxis resistant to treatment

- β-blockade blunts the β1 (inotropic/chronotropic) and β2 (bronchodilator, mediator-release inhibition) effects of adrenaline
- Unopposed α effects and reflex vagal tone may cause bradycardia
- β-blocked patients have more severe and refractory reactions
- ACE inhibitors also worsen severity (bradykinin)

Which drug would you give next? Explain its mechanism and dose

- Glucagon
- Activates adenylate cyclase via glucagon receptors, raising cAMP independently of β-receptors → inotropy and chronotropy
- Dose 1–5 mg IV over 5 minutes, then infusion 5–15 µg/min titrated
- Side effects: nausea and vomiting (protect airway), hyperglycaemia

Anaphylaxis on beta-blockers — Q13 (cont.)

Outline the management of refractory anaphylaxis

- Adrenaline infusion titrated (start 0.05–0.1 µg/kg/min), via dedicated line
- Rapid fluids 500–1000 mL boluses (large volumes may be needed – capillary leak)
- Second vasopressor: noradrenaline or vasopressin; consider methylene blue for vasoplegia
- Airway: early intubation by the most experienced person if stridor/oedema; nebulised adrenaline
- Antihistamines and steroids are adjuncts only – they do not treat shock
- Send mast cell tryptase (soon after, 1–2 h, and > 24 h); observe for biphasic reaction
- ECMO in refractory shock/arrest; atropine for bradycardia

Anaphylaxis on beta-blockers — Q14

List the most common causes of perioperative anaphylaxis and grade the severity

- Common triggers: antibiotics (now the leading cause in UK, e.g., teicoplanin, cefazolin), NMBAs (rocuronium, suxamethonium), chlorhexidine, patent blue dye
- Latex less common now
- Grade I skin only; II moderate multi-organ; III life-threatening (shock, severe bronchospasm); IV cardiac/respiratory arrest
- This is grade III

Describe the immediate management

- Call for help, stop all trigger agents, 100% O2, secure airway
- Adrenaline IV 50 µg boluses (titrated) in anaesthetised patient (IM 0.5 mg if no IV access/monitoring)
- Rapid crystalloid 20 mL/kg; elevate legs
- Start adrenaline infusion if more than 2–3 boluses needed
- CPR with standard arrest doses if SBP < 50 mmHg or arrest

What investigations and follow-up are needed

- Mast cell tryptase: as soon as possible after resuscitation, at 1–2 h, and baseline at > 24 h
- Raised tryptase supports IgE/mast cell mechanism (normal does not exclude)
- Document all drugs and timings; report to national system
- Refer to allergy clinic for skin testing (4–6 weeks later)
- Inform patient and GP; alert bracelet

Post-ROSC care and temperature control — Q15

Describe the ERC–ESICM 2021 recommendations for temperature management and the evidence behind them

- Actively prevent fever (> 37.7 °C) for at least 72 h in patients who remain comatose
- TTM2 (2021): 33 °C vs normothermia with early fever treatment – no difference in 6-month mortality or function
- TTM1 (2013): 33 vs 36 °C – no difference
- Do not actively rewarm hypothermic patients quickly; avoid shivering (sedation, magnesium)

Outline your physiological targets in the first 24 hours

- MAP > 65 mmHg (and adequate urine output, falling lactate)
- SpO2 94–98%; avoid hypoxaemia and hyperoxia
- Normocapnia (PaCO2 35–45 mmHg)
- Glucose 7.8–10 mmol/L; avoid hypoglycaemia
- Treat seizures; continuous or intermittent EEG
- Lung-protective ventilation 6–8 mL/kg PBW

When is urgent coronary angiography indicated

- STEMI on ECG → emergency angiography/PCI
- No ST elevation: angiography only if high suspicion of ongoing coronary occlusion or haemodynamic/electrical instability
- COACT and TOMAHAWK: no benefit from immediate routine angiography in non-STEMI post-arrest

Post-ROSC care and temperature control — Q16

Describe the ERC–ESICM multimodal neuroprognostication algorithm

- Assess at ≥ 72 h after ROSC, after excluding confounders (sedation, hypothermia, metabolic)
- Prerequisite: comatose with motor response ≤ extension (M ≤ 3)
- Poor outcome likely if ≥ 2 of: absent pupillary and corneal reflexes at ≥ 72 h; bilaterally absent N20 SSEP waves at ≥ 24 h
- Highly malignant EEG (suppressed background ± periodic discharges, burst suppression) at > 24 h
- NSE > 60 µg/L at 48 h and/or 72 h; status myoclonus ≤ 72 h
- Diffuse anoxic injury on CT (loss of grey–white differentiation) or MRI (DWI changes)

What confounders must be excluded

- Residual sedatives/analgesics/neuromuscular blockers (renal/hepatic failure prolongs)
- Hypothermia
- Severe metabolic derangement (Na, glucose, urea, hepatic encephalopathy)
- Non-convulsive status epilepticus

How would you communicate uncertainty to the family

- Private setting, senior clinician with nurse; use plain language
- Explain that prognosis is based on several tests over time, not one
- Avoid premature withdrawal decisions (self-fulfilling prophecy)
- Document, offer follow-up meetings; consider organ donation pathway if appropriate

Hyperkalaemic cardiac arrest — Q17

List the 'reversible causes' of cardiac arrest and identify the likely one here

- 4 Hs: hypoxia, hypovolaemia, hypo/hyperkalaemia and metabolic, hypothermia
- 4 Ts: thrombosis (coronary/pulmonary), tension pneumothorax, tamponade, toxins
- Likely hyperkalaemia (missed dialysis, peaked T waves, K+ 8.1)

Outline the specific treatment during CPR, with doses

- Calcium chloride 10% 10 mL IV rapid bolus (repeat if needed) – membrane stabilisation
- Insulin 10 units soluble + 25 g glucose IV rapid injection
- Sodium bicarbonate 8.4% 50 mL IV if severe acidosis or renal failure
- Consider haemodialysis during CPR for refractory arrest; ECPR in selected centres
- Continue standard ALS; avoid treating with magnesium or amiodarone instead

What should be done after ROSC

- Monitor glucose (hypoglycaemia after insulin) hourly for 6 h; glucose infusion
- Check K+ frequently (rebound rise as insulin effect wanes)
- Urgent dialysis
- Potassium binder (sodium zirconium cyclosilicate or patiromer); salbutamol nebs as adjunct

Board 2026 recall · Hyperkalaemic cardiac arrest · Lott C et al. ERC Guidelines 2021: Cardiac arrest in special circumstances. Resuscitation 2021

Hyperkalaemic cardiac arrest — Q18

Describe the ECG changes of hyperkalaemia in order of severity

- Peaked (tented) T waves
- PR prolongation, P-wave flattening/loss
- QRS widening, bradycardia, junctional rhythm
- Sine wave pattern → VF/asystole/PEA

Explain how calcium works and describe the emergency treatment

- Calcium raises the threshold potential, restoring the gap with the resting membrane potential → stabilises the myocardium (does not lower K+)
- Calcium gluconate 10% 30 mL (or chloride 10 mL) over 5–10 min; repeat if ECG unchanged; works within minutes, lasts 30–60 min
- Shift: insulin 10 U + glucose 25 g; salbutamol 10–20 mg nebulised
- Bicarbonate only if acidotic
- Remove: stop ACEI/spironolactone, binders, loop diuretic if passing urine, dialysis if refractory/anuric
- Transcutaneous pacing / isoprenaline for bradycardia if unresponsive

List the causes of pseudohyperkalaemia and of true hyperkalaemia

- Pseudo: haemolysed sample, prolonged tourniquet/fist clenching, very high WCC or platelets, delayed processing, EDTA contamination
- True – reduced excretion: AKI/CKD, drugs (ACEI/ARB, spironolactone, trimethoprim), Addison, type 4 RTA
- True – shift/release: acidosis, rhabdomyolysis, tumour lysis, suxamethonium, insulin deficiency, digoxin toxicity

Fluids in resuscitation — Q19

Calculate the anion gap and explain the acid–base disorder

- AG = Na − (Cl + HCO3) = 141 − (118 + 16) = 7 mmol/L (corrected for albumin ≈ 9.5) → normal
- Hyperchloraemic normal anion gap metabolic acidosis from large-volume saline
- Saline has Na = Cl = 154 mmol/L, SID = 0 → dilutes plasma SID → acidosis
- Respiratory compensation appropriate (expected PaCO2 = 1.5 × 16 + 8 ± 2 = 30–34; actual 36 → mild additional respiratory acidosis)

Which fluid would you choose and why? Summarise the evidence

- Balanced crystalloid (Hartmann's, Plasma-Lyte) preferred in sepsis (SSC 2021)
- SMART (2018): balanced fluids reduced major adverse kidney events vs saline
- BaSICS and PLUS (2021–22): no significant mortality difference overall; possible harm of saline in TBI excluded – use saline in brain injury
- Avoid starches (6S, CHEST: AKI, RRT, mortality) and gelatins
- Albumin may be added after large crystalloid volumes (ALBIOS: no overall benefit, possible benefit in septic shock)

What are the harms of fluid overload in sepsis

- Pulmonary oedema, worse oxygenation, longer ventilation
- AKI (renal venous congestion), raised intra-abdominal pressure
- Gut oedema, ileus, poor wound healing
- Positive fluid balance independently associated with mortality; consider de-resuscitation once stable (ROSE model)

Fluids in resuscitation — Q20

Define fluid responsiveness and describe how you would assess it in this patient

- Increase in stroke volume or cardiac output ≥ 10–15% after fluid challenge
- Passive leg raise with real-time CO measurement (VTI, pulse contour) – increase ≥ 10% predicts response
- Pulse pressure variation > 12–13% (only if fully ventilated, VT ≥ 8 mL/kg, sinus rhythm, closed chest)
- Mini-fluid challenge (100–250 mL) with VTI change; end-expiratory occlusion test
- IVC variation and CVP are poor predictors

What are the limitations of using pulse pressure variation

- Spontaneous breathing efforts
- Arrhythmias (AF)
- Low tidal volume (< 8 mL/kg) or low lung compliance → false negatives
- Open chest, RV failure, raised intra-abdominal pressure → false positives/negatives

Describe the phases of fluid therapy (ROSE model)

- Resuscitation: rapid boluses to restore perfusion (minutes)
- Optimisation: titrated fluid guided by responsiveness (hours)
- Stabilisation: maintenance only, aim neutral balance (days)
- Evacuation (de-resuscitation): active fluid removal with diuretics/ultrafiltration